

SAI DHRUV YELLANKI

College Park, Maryland (**Ready to Relocate**)

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Professional Summary

AI Engineer with 2+ years building production-grade Generative AI and predictive ML systems across enterprise proposal automation, healthcare, supply chain, and financial domains on Azure and AWS. Specialized in LLM fine-tuning (LoRA, RLHF), GraphRAG pipelines, and multi-agent orchestration with LangGraph and LangChain. Proven deploying containerized systems using Docker and Kubernetes with CI/CD in HIPAA, GDPR, and ISO 27001 compliant environments.

Education

University of Maryland

Master of Science in Data Science; GPA: 3.80/4.0

August 2024 – May 2026

College Park, MD, USA

SRM Institute of Science and Technology

Bachelor of Technology in Computer Science and Business Systems ; GPA: 3.74/4.0

June 2019 – May 2023

Chennai, TN, India

Experience

Aurelius Tech & Talent Solutions

AI Engineer

June 2025 – Present

Ellicott City, MD, USA

- Architected a **GraphRAG** pipeline on Fabric **OneLake**, Neo4j, and PostgreSQL/pgvector integrating HNSW vector search, BM25, multi-hop Cypher graph traversal, and **RRF fusion**, cutting research time by **65%** across knowledge-intensive workflows.
- Engineered an **ms-marco-MiniLM cross-encoder reranking** layer within the GraphRAG pipeline, improving document retrieval precision by **38%** and surfacing source-attributed answers at **sub-500ms** retrieval latency.
- Fine-tuned **gpt-oss-20b** as the reasoning backbone for a 38-agent knowledge-intensive workflow system via Azure AI Agent Service and Semantic Kernel, achieving **81% F1** on **RAGAS** faithfulness evals and a **14% accuracy gain** over human baselines.
- Replaced manual **prompt engineering** with a **DSPy** programmatic signature layer across the gpt-oss-20b agent ensemble, enforcing output structure and improving answer reasoning precision by **19%** over prompted baselines. .
- Built a persistent agent memory layer using **Neo4j** Context graphs, PostgreSQL audit trails, and **OpenMemory** long-term recall, reducing **hallucination rates** by **22%** across multi-turn knowledge-intensive workflow sessions.
- Deployed GraphRAG and agent inference services on Azure Container Apps and **AKS** with SAST, SCA, and secret scanning in **ISO-compliant CI/CD** and **OpenTelemetry** observability, cutting release cycles by **40%** and sustaining **99.6% uptime**.

Watzmann Consulting

Data Analyst – Predictive Analytics & Machine Learning

June 2023 – June 2024

Hyderabad, TG, India

- Constructed **ETL pipelines** on **Azure Databricks** using **PySpark** and **SQL** to ingest 200K+ records from **Azure Data LakeStorage Gen2**, implementing a Delta Lake Architecture for structured, versioned features for all downstream models.
- Developed a Bi-LSTM and XGBoost demand **forecasting ensemble** improving prediction **accuracy by 12%** and a MILP-paired workload optimization system to **minimize high compute execution** intervals, reducing stock-outs and delivering 15% ROI.
- Executed live **A/B testing** to compare the Bi-LSTM ensemble against legacy ARIMA and Prophet models, providing the empirical validation required to prove a 15% lift in operational efficiency before production rollout.
- Implemented a production **MLOps** lifecycle using the **MLflow** Model Registry to automate retraining and versioning, integrating interactive **Power BI** dashboards to monitor real-time model-drift and reduce deployment cycles by 35%.
- Pioneered a **RAG application** using **LangChain**, **ChromaDB**, and Llama 2 to automate Root Cause Analysis of forecasting anomalies, enabling managers to query inventory drivers in natural language and reducing investigation time by 45%.

Projects

LoRA Tuned RLHF News Summarization System | *RLHF, RAG, LORA, Langchain, Docker, vLLM* | [GitHub Link](#)

- Deployed a news-summarization platform using **RLHF (PPO/trlx)** to **fine-tune a T5 model** integrated with a **pgvector RAG pipeline**, improving ROUGE-L by 11% and achieving 240ms p95 latency using **vLLM**.

BioGraphX: Multi-Agent Biomedical RAG System | *Python, LangGraph ,RAG, Qwen2.5, ChromaDB, Docker* | [GitHub Link](#)

- Architected a biomedical QA system **orchestrating 7 LangGraph agents** and a **LoRA fine-tuned Qwen2.5 model with Chain-of-Thought prompting**, achieving 96% ROUGE-L improvement across 298K PubMed records with sub-8s inference using Docker/FastAPI deployment.

Technical Skills

Programming Languages & Databases: Python (Pandas,Scikit-learn, PySpark, PyTorch) SQL, R, SurrealDB, PostgreSQL.

Tools & Technologies: Docker, Kubernetes, Azure (Machine Learning, Azure Data Lake, Databricks, Fabric, AKS, Onelake), AWS(Bedrock, Sagemaker), Git, Power BI, Tableau, Grafana, CI/CD Pipelines, Microsoft Copilot, Microsoft Power Platform .

Machine Learning & Analytics: Predictive Modeling , Time Series Forecasting , Deep Learning , NLP , A/B Testing.

AI & LLM Engineering:Multi-Agent Orchestration (LangGraph, LangChain, AutoGen), GraphRAG, DSPy (Programmatic Logic), Fine-tuning (LoRA, QLoRA, RLHF), Claude code (skills creation, MCP (creating and connecting) , debugging),Prompt Engineering, Long-term Memory (OpenMemory), vLLM Inference.